

MTH 207 Practice Exam 1 (Dr. Kim, Fall 2026)

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Rules

- The cover page should only have your name and email. Nothing else may may be written on the cover page.
- No calculators.
- No notes, books, or other outside resources.
- No phones, smartwatches, earbuds, or any other device that communicates.
- Eyes on your own exam.
- Scratch paper is available upon request. You may only use the provided scratch paper, and all scratch paper (used and unused) must be submitted with your exam.

Expectations

- Write with proper notation.
- Show your work in order to earn points. University mathematics focuses on the **process** more than the **answer**. For example, stating “does not exist” without giving a reason will not earn points.
- *There is no guarantee that the practice exam has exactly all the same types of problems as the actual exam. (However, the practice exam is a good study resource for the actual exam.)*

1. Find $\lim_{x \rightarrow 3} \frac{8}{|x - 3|}$

2. Find $\lim_{p \rightarrow 2} \frac{p^2 - 4}{p - 2}$

3. Find $\lim_{x \rightarrow 0} \frac{x}{|x|}$

4. Find $\lim_{x \rightarrow 0} x^2 \sin(\frac{1}{x})$

5. Find $\lim_{x \rightarrow -\infty} \frac{4x^5 + 6}{7x^8 + 9}$

6. Find $\lim_{x \rightarrow \frac{4\pi}{3}} \cos(x)$

7. Find $\lim_{x \rightarrow 2} \frac{x+6}{x+3}$

8. Find $\lim_{c \rightarrow 0} \frac{c^2}{\sqrt{c^2 + 100} - 10}$

9. Let $f(x) = x^2 + 2x + \sin(\pi x)$. Prove that there exists a number c in $(4, 10)$ such that $f(c) = 48.75$. *Optional reading in footnote¹.*

¹If $f(x)$ represents the temperature of an object x hours after midnight, the work completed proves that some time between 4 hours after midnight and 10 hours after midnight, the object had a temperature of 48.75 degrees.

10. Given $f(x) = x^2 + 10x$ find $f'(3)$ directly by a limit-based definition. (No credit will be given for applying derivative rules such as the Power Rule.) *Optional reading in footnote².*

11. Find r and s so that

$$f(x) = \begin{cases} x^2 - 2r + 1 & \text{if } x < 5 \\ 3x + s & \text{if } x = 5 \\ 6r - 2x - 12 & \text{if } x > 5 \end{cases}$$

is continuous at $x = 5$, applying the definition.

12. Find a horizontal asymptote of $f(x) = \frac{5x^2 + 8}{6x^2 - 1}$

²If $f(x)$ represents the position (in meters) of a boat x minutes after midnight, then $f'(3)$ finds the velocity of the boat (in meters per minute) 3 minutes after midnight.

